ADS401 Evaluation Methods and Statistics

Presentation 2 Supplementary Materials

Usability Evaluation of the XJTLU Museum AI System

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This is one combined PDF containing participant materials, task materials, measures, a blank data template, and an implementation reflection.

01 / STUDY DESIGN SUMMARY

Study design summary

Research questions
RQ1: Can visitors complete the four core Museum AI tasks?
RQ2: How usable and efficient is the system?
RQ3: What usability barriers do participants report?

Target users
XJTLU students or staff aged 18+, with mixed digital confidence and museum experience.

Design

Moderated task-based usability test with a single system, followed by a 10-question bilingual Wenjuanxing survey and a short exit interview. The study evaluates one system version and does not compare separate conditions.

Sample and data

12 to 15 participants. Measures include independent task success, time, errors, assistance, SUS, task ease, task confidence, content relevance, navigation, recovery, trust, cognitive load, and think-aloud notes.

Hypotheses

Hypothesis	Prediction	Evidence
H1	At least 80% independent success on each task	Observation sheet
H2	SUS reaches the benchmark of 68	SUS items in Wenjuanxing
H3	Ease and confidence are positively related to SUS	Task ratings and SUS
H4	Onboarding, navigation, answer clarity, or recovery barriers recur	Open feedback and think-aloud

The design evaluates one system. Results will describe the tested version, tasks, and participant group, not the wider population.

02 / INFORMATION SHEET

Information sheet

Study title

Usability Evaluation of the XJTLU Museum AI System

Invitation

You are invited to take part in a short usability study. The study evaluates how easy it is for visitors to use the XJTLU Museum AI system during a museum tour.

What you will do

You will complete four short tasks with the Museum AI system. The session should take about 15 minutes. You will think aloud, complete a 10-question bilingual Wenjuanxing survey, and answer a few exit questions.

Participant survey

Live link: https://v.wjx.cn/vm/QZpCC3h.aspx. The survey uses an anonymous participant code and does not ask for a name or student number.

What we record

We record task success, completion time, errors, moderator assistance, questionnaire responses, and short interaction notes. We do not record names, student numbers, passwords, photographs, or personal museum records.

Risks and benefits

There are no expected physical risks. You may experience mild frustration if a task is difficult. You can stop at any time. The study will help identify practical improvements to the Museum AI system.

Voluntary participation

Participation is voluntary. You may decline a question or withdraw before analysis without giving a reason and without penalty.

Data use and storage

Data will be anonymised and stored in a protected course workspace. Only the group and authorised module staff will access anonymous data. We will not publish identifiable material.

Contact

Group H: Cheng Qian, Chen Cui, Ziqian Wang, Simu He. Module: ADS401 Evaluation Methods and Statistics.

03 / CONSENT FORM

Consent form

Participant ID: _____ Date: _____

Please read each statement and mark [X] if you agree.

[ ] I have read and understood the information sheet.

[ ] I had the opportunity to ask questions.

[ ] I understand that participation is voluntary.

[ ] I understand that I may stop at any time.

[ ] I agree to anonymous task and questionnaire data being used for this coursework.

[ ] I understand that no identifying information will be published.

Participant statement
I agree to take part in the study described above.

Participant signature or typed initials: _____

Date: _____

If the session is online, typing your initials can serve as consent. The researcher will save a dated copy.

04 / RECRUITMENT AND SCREENING

Recruitment and screening

Short recruitment message
We are recruiting 12-15 XJTLU students or staff for a 15-minute study of the XJTLU Museum AI system. You will complete four short museum-tour tasks and answer a few questions about usability. No prior experience with the system is needed. Participation is anonymous and voluntary.

Screening questions

Are you 18 or older? _____

Are you an XJTLU student or staff member? _____

Have you used a museum AI guide before? _____

Have you visited XJTLU Museum before? _____

Recruitment record

Participant ID	Channel	Eligible	Session date
P01			
P02			
P03			

Channel examples: approved course announcement, student society, snowball referral.

05 / MODERATOR SCRIPT

Moderator script

Opening
Thank you for coming. Today we are testing the Museum AI system, not you. Please think aloud as you work. I may stay quiet while you complete each task. You can stop at any time.

Before the tasks
This is the XJTLU Museum context. The AI system is intended to help visitors find exhibit information and navigate the tour. Please explore it as you normally would.

Task instructions

Read each task from the task card. Do not explain where to click. If the participant stops for 20-30 seconds, ask: What are you thinking right now?

Assistance rule

Provide help only if the participant cannot continue. Record the point of assistance, the type of help, and whether the task was completed.

After the tasks

Please complete this short usability questionnaire. There are no right or wrong answers. Then I will ask about what was easy or difficult.

Avoid praising or correcting the participant during the tasks. Keep the same tone and wording across sessions.

06 / TASK CARDS

Task cards

Task 1: Start the guide
Open the XJTLU Museum AI system and select one exhibit to begin the tour.
Success: an exhibit-specific screen without help.

Task 2: Ask about an exhibit
Ask one question about the selected exhibit and identify the key information in the response.
Success: a relevant question and a stated answer.

Task 3: Move to the next stop
Use the system to find the next recommended exhibit or tour stop.
Success: identify or reach the next stop without help.

Task 4: Recover from an unclear answer
The moderator provides an unclear-response scenario. Ask again or use another method to resolve it.
Success: a reasonable recovery action and explanation.

07 / OBSERVATION SHEET

Observation sheet

Participant ID: _____ Date: _____ Moderator: _____

Task	Outcome	Time	Errors / help	Notes
1 Start	Independent / assisted / failed	____ sec		
2 Ask	Independent / assisted / failed	____ sec		
3 Navigate	Independent / assisted / failed	____ sec		
4 Recover	Independent / assisted / failed	____ sec		

Repeated problems

Record repeated hesitation, wrong turns, unclear labels, or workaround behaviour.

Moderator reflection

Was the task wording clear? Did the system behave consistently? What should change before the next session?

08 / QUESTIONNAIRE FLOW AND SUS

Questionnaire flow and SUS

Live Wenjuanxing survey

Participant link: https://v.wjx.cn/vm/QZpCC3h.aspx
Researcher dashboard: https://www.wjx.cn/report/393565465.aspx?sat=1

Ten main questions

No.	Content
1	Role at XJTLU
2	Prior use of the Museum AI system
3	Completion matrix for four tasks
4	Task ease for four tasks, 1-7
5	Task confidence for four tasks, 1-7
6	Start clarity, content relevance, navigation, recovery, trust, cognitive load
7	Standard SUS, 10 items
8	Main obstacle
9	Useful, unclear, or missing exhibit information
10	Most important content or interaction improvement

The printed SUS block below documents the exact items used in Question 7 and supports scoring checks.

1 = strongly disagree, 5 = strongly agree

Item	1	2	3	4	5
1. I think that I would like to use this system frequently.					
2. I found the system unnecessarily complex.					
3. I thought the system was easy to use.					
4. I think that I would need technical support to use this system.					
5. I found the various functions in this system were well integrated.					
6. I thought there was too much inconsistency in this system.					
7. I would imagine that most people would learn to use this system quickly.					
8. I found the system very cumbersome to use.					
9. I felt very confident using the system.					
10. I needed to learn a lot before I could get going.					

Scoring: for odd items use score - 1. For even items use 5 - score. Sum the ten adjusted scores and multiply by 2.5.

09 / OPEN FEEDBACK AND EXIT QUESTIONS

Open feedback and exit questions

Survey open questions

Question 8 records the main obstacle. Question 9 asks what exhibit information was useful, unclear, or missing. Question 10 asks for the most important content or interaction improvement.

Exit interview prompts

Use only neutral prompts: What were you thinking at that point? What made that step easy or difficult? What would help a first-time visitor? Did you trust the answer, and why?

Open questions

What was easiest about the system?

Where did you hesitate or feel unsure?

What would help a first-time visitor?

Was there anything you did not trust or understand?

What is the one change you would make first?

Notes

> Record key phrases and repeated ideas. Do not include names or identifiable details.

10 / DATA TEMPLATE AND ANALYSIS

Data template and analysis

Participant-level data structure

ID	Task outcome	Time	Ease	Confidence	SUS	Trust	Theme
P01							
P02							
P03							

Coding rules

Variable	Code	Definition
Task outcome	1 / 0	Independent success / assisted or failed
Time	Seconds	Instruction to completion or stop
SUS	0-100	Standard alternating-item scoring
Ease and confidence	1-7	Higher is more positive
Theme	Text	Onboarding, navigation, clarity, recovery, other

Planned analysis

Report success rate and errors by task; median and range for time; SUS with a benchmark test; Cronbach's alpha for the ten SUS items; exploratory Spearman correlations for H3; and a theme table with anonymised examples for H4. Check normality, missingness, duplicate IDs, and outliers before choosing the final test.

11 / PILOT AND IMPLEMENTATION LOG

Pilot and implementation log

System version: _____ Access method: _____

Pilot date: _____ Pilot participants: _____

Session duration: _____ Moderator: _____

Fallback used: _____ Changes made: _____

What changed after the pilot

Issue observed	Change made	Reason

Implementation reflection

> Record access constraints, technical problems, task wording changes, timing, and deviations from the plan. Do not report data that were not collected.

12 / ETHICS AND AI-USE STATEMENT

Ethics and AI-use statement

Ethics checklist

[ ] Ethics approval completed before recruitment.

[ ] Information sheet and consent form used.

[ ] No passwords or personal accounts requested.

[ ] No photographs of participants collected.

[ ] Anonymous participant IDs used.

[ ] Raw identifiable material excluded from public files.

AI-use statement

> [Complete accurately before submission.] The group used AI to support document structure and formatting. The module permits only non-substantive AI use. The group must verify and rewrite substantive academic content and state the actual tools and purposes used. Do not claim a narrower use than actually occurred.

Material acknowledgement

The evaluation concerns the XJTLU Museum AI-supported visitor system. No ownership of the original system, Museum collections, or institutional materials is claimed.

Before presenting, run the two-person pilot, update the pilot record, and replace pending-data placeholders with information actually collected. Do not present simulated results as participant evidence.